Monte Carlo Simulations of the 2D Ising Model

Assignment 2: Many Body Physics

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1 Introduction

The Ising model stands as a paradigmatic system in statistical mechanics, serving as a minimal model for ferromagnetism and phase transitions. Despite its discrete nature and simple interactions, it captures the essential physics of symmetry breaking and critical phenomena.

In this assignment, we perform a comprehensive numerical study of the two-dimensional Ising model on a square lattice. Our primary objectives are to:

1. Simulate the system using Markov Chain Monte Carlo (*MCMC*) methods.
2. Compare the efficiency of local update algorithms (*Metropolis*) against global cluster algorithms (*Wolff/Swendsen-Wang*) in the context of critical slowing down.
3. Estimate the critical temperature T_c through the analysis of magnetization, specific heat, and autocorrelation times.
4. Explore advanced extensions including the XY Model and unsupervised phase detection using Principal Component Analysis (*PCA*).

2 Theoretical Framework

2.1 Model Definition

We consider a system of $N = L \times L$ spins on a square lattice, where each spin variable σ_i can take values $\{-1, +1\}$. The system is governed by the Hamiltonian:

$$H(\{\sigma\}) = -J \sum_{\langle i,j \rangle} \sigma_i \sigma_j \tag{1}$$

where $J > 0$ represents the ferromagnetic coupling constant and the summation $\langle i, j \rangle$ extends over nearest-neighbor pairs. We employ periodic boundary conditions to mitigate surface effects and approximate the thermodynamic limit.

2.2 Observables

To characterize the phase transition, we measure the following thermodynamic quantities:

- **Magnetization per spin (M):** The order parameter of the system.

$$M = \frac{1}{N} \left| \sum_{i=1}^{N} \sigma_i \right|$$

- **Specific Heat (C_V):** Derived from energy fluctuations.

$$C_V = \frac{\langle E^2 \rangle - \langle E \rangle^2}{N k_B T^2}$$

3 Methodology

3.1 Initialization and Visualization

The system is initialized with a random configuration where each site is assigned spin up $(+1)$ or down (-1) with equal probability. This *"hot start"* corresponds to infinite temperature conditions.

Figure 1: Visualization of the 2D Ising lattice initialization. Light and dark pixels represent +1 and −1 spins respectively.

3.2 Simulation Algorithms

3.2.1 Single Spin Flip (Metropolis)

The standard Metropolis algorithm proposes a flip $\sigma_i \to -\sigma_i$ and accepts it with probability $A = \min(1, e^{-\beta \Delta E})$. While ergodic, this local update method suffers from *critical slowing down* near T_c, where the correlation length ξ diverges.

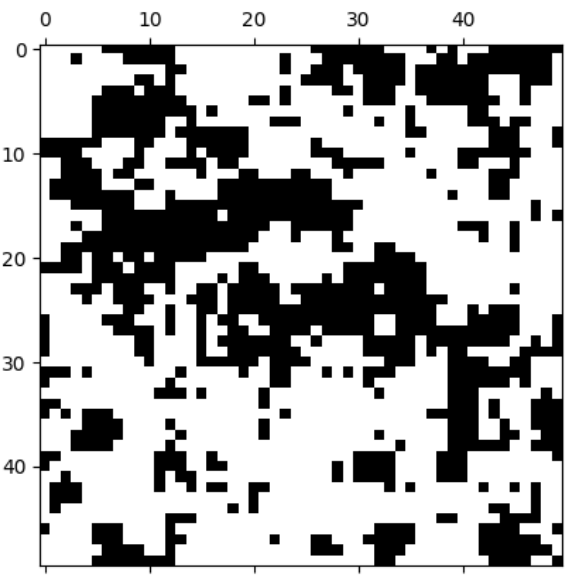


Figure 2: Metropolis simulation animation

3.2.2 Cluster Algorithms (Wolff/Swendsen-Wang)

To accelerate equilibration, we implemented the **Wolff cluster algorithm**. This rejection-free method constructs clusters of aligned spins based on the bond probability $P_{add} = 1 - e^{-2\beta J}$ and flips them collectively. This global update mechanism significantly reduces the dynamic critical exponent z.

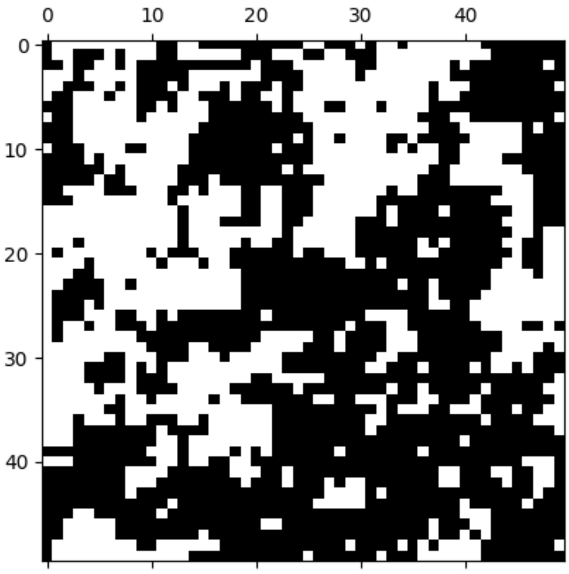


Figure 3: Wolff simulation animation

3.3 Error Analysis: Binning

To account for temporal correlations in the MCMC time series, we employed the binning method. Measurements were grouped into blocks of size K, and the variance of block averages was analyzed to estimate the true autocorrelation time τ and the standard error.

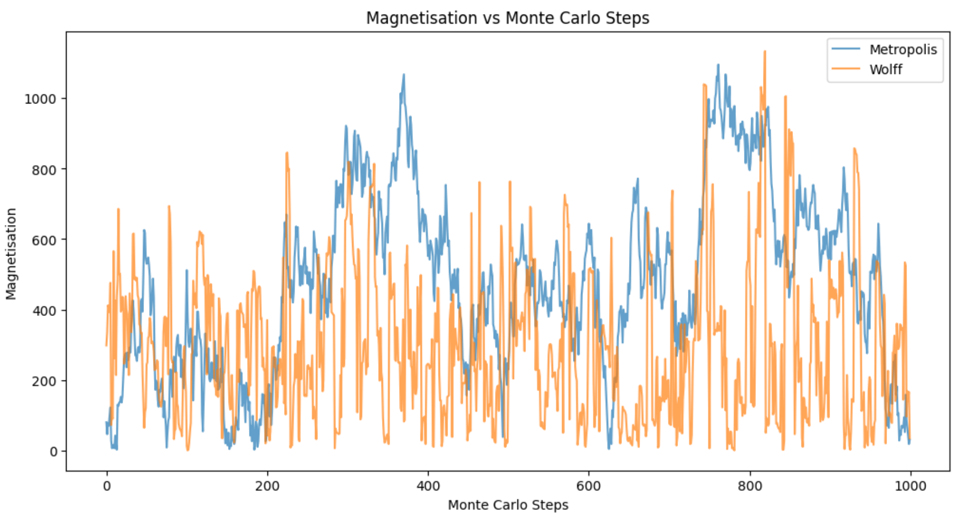
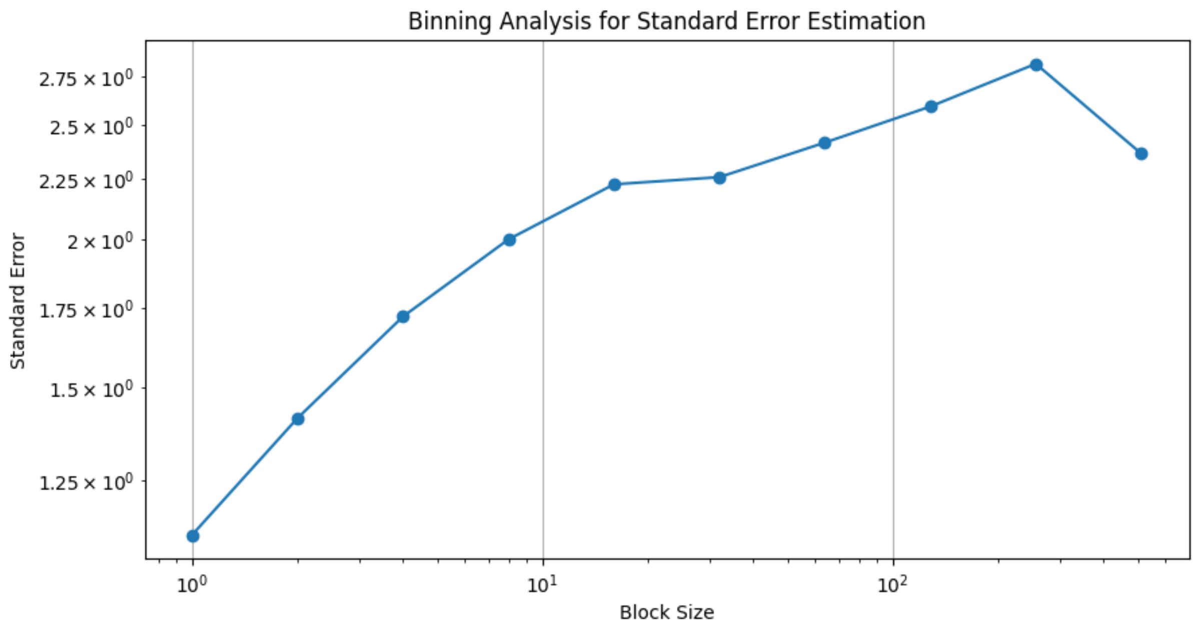


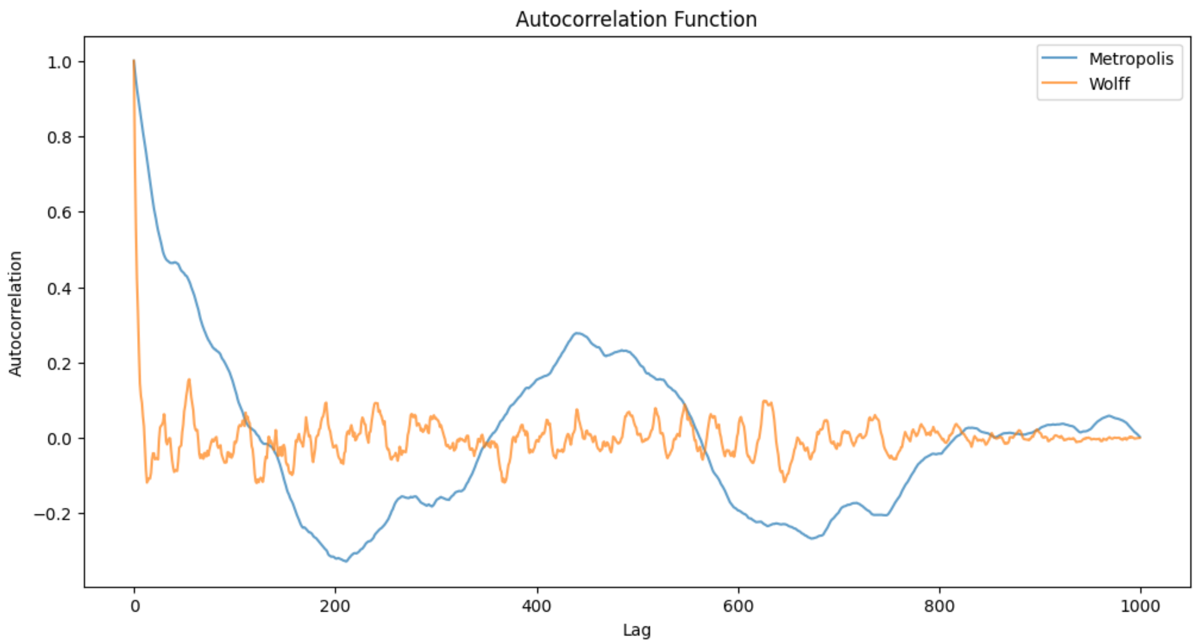
Figure 4: Comparison of Magnetization vs. Iterations for SSF and Wolff algorithms. The cluster algorithm (Wolff) rapidly reaches the ground state magnetization ($|M| \approx 1$).



4 Results and Discussion

4.1 Convergence Comparison

We compared the convergence rates of the Single Spin Flip (SSF) and Wolff algorithms at low temperatures. As illustrated in Figure 5, the Wolff algorithm equilibrates orders of magnitude faster than SSF, validating its efficacy in overcoming energy barriers.
Integrated Autocorrelation Time (Metropolis): *9.88860124646597*
Integrated Autocorrelation Time (Wolff): *2.9510518130070866*
Estimated Error (Metropolis): *20.490649672734143*
Estimated Error (Wolff): *19.726901635592753*



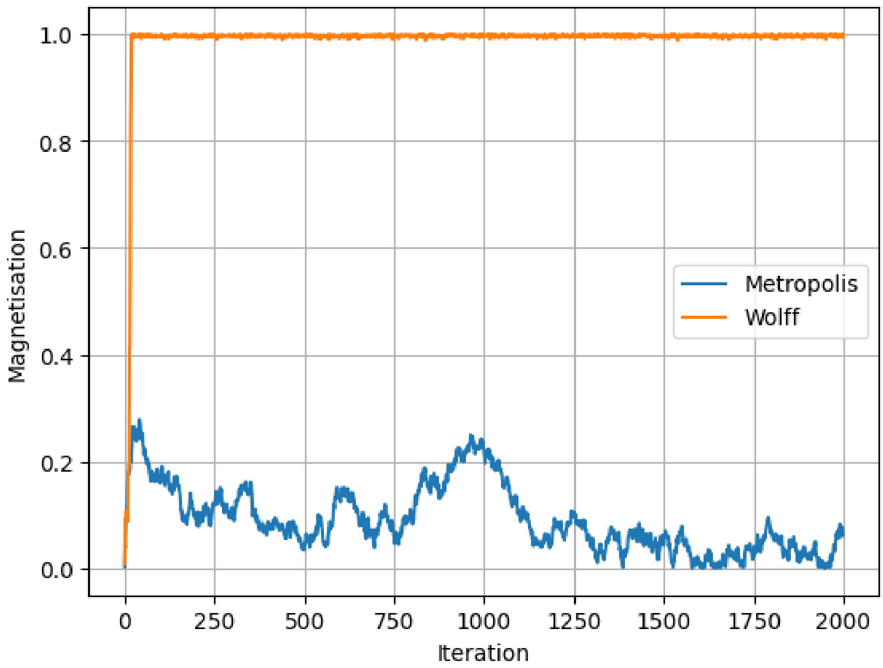


Figure 5: Comparison of Magnetization vs. Iterations for SSF and Wolff algorithms. The cluster algorithm (Wolff) rapidly reaches the ground state magnetization ($|M| \approx 1$).

4.2 Phase Transition Analysis

4.2.1 Magnetization

The magnetization was simulated for lattice sizes $L \in \{8, 10, 12\}$ over a temperature range bracketing the critical point. Figure **??** displays the spontaneous symmetry breaking.

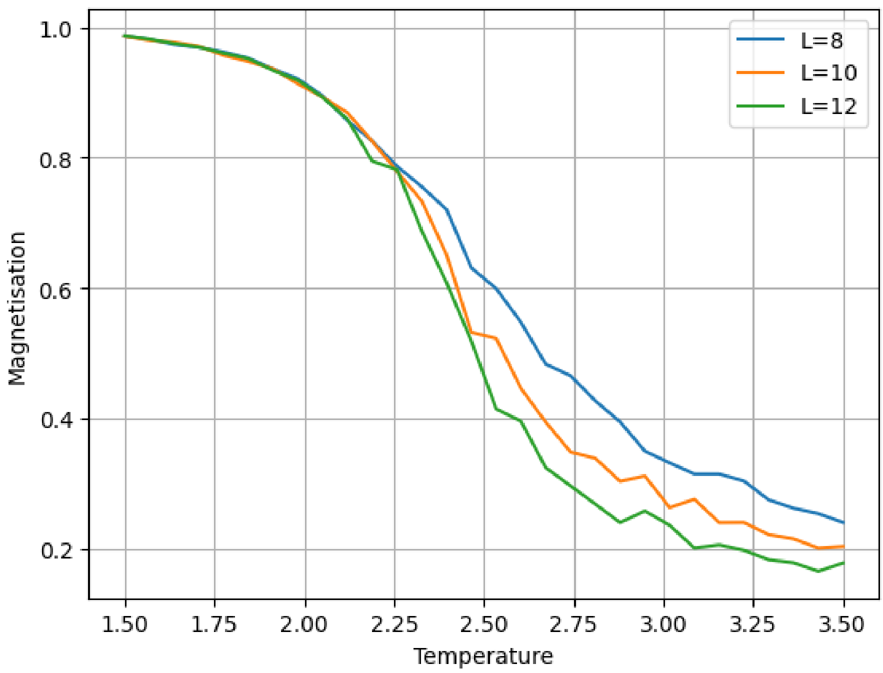
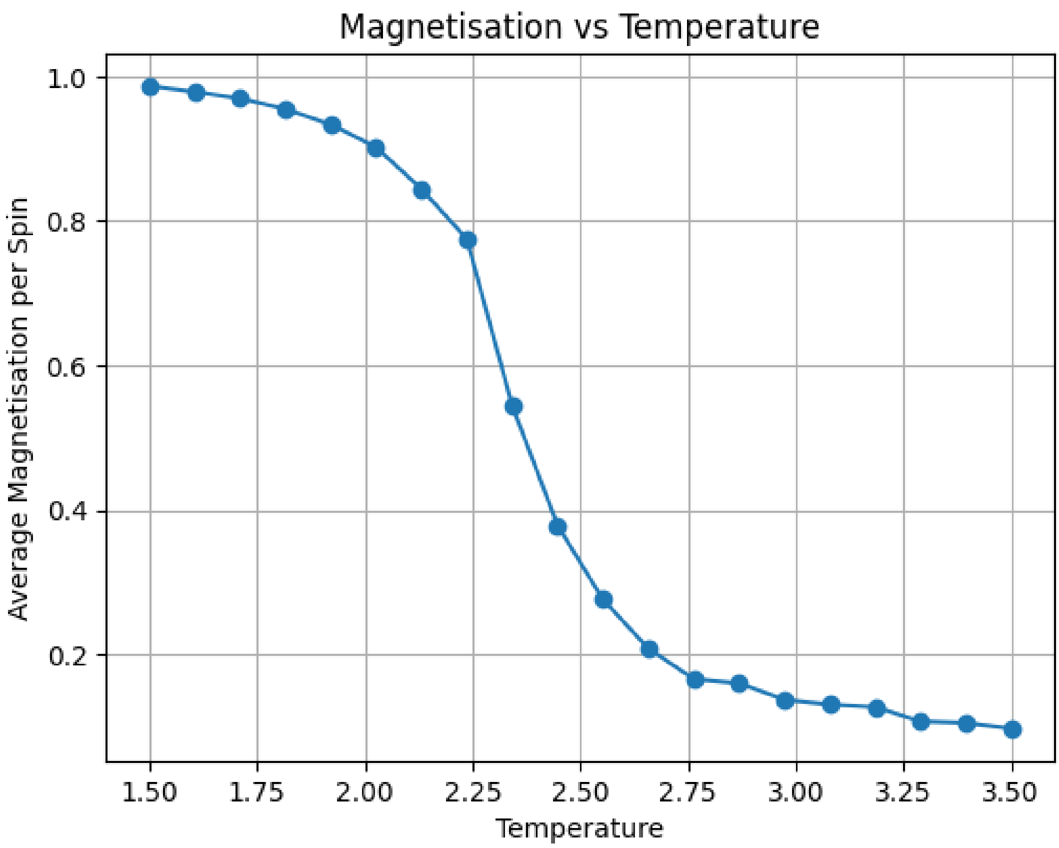


Figure 6: Magnetization ($|M|$) as a function of Temperature (T) for different system sizes. The transition sharpens as L increases.



4.2.2 Autocorrelation Time

We analyzed the integrated autocorrelation time τ as a function of temperature.

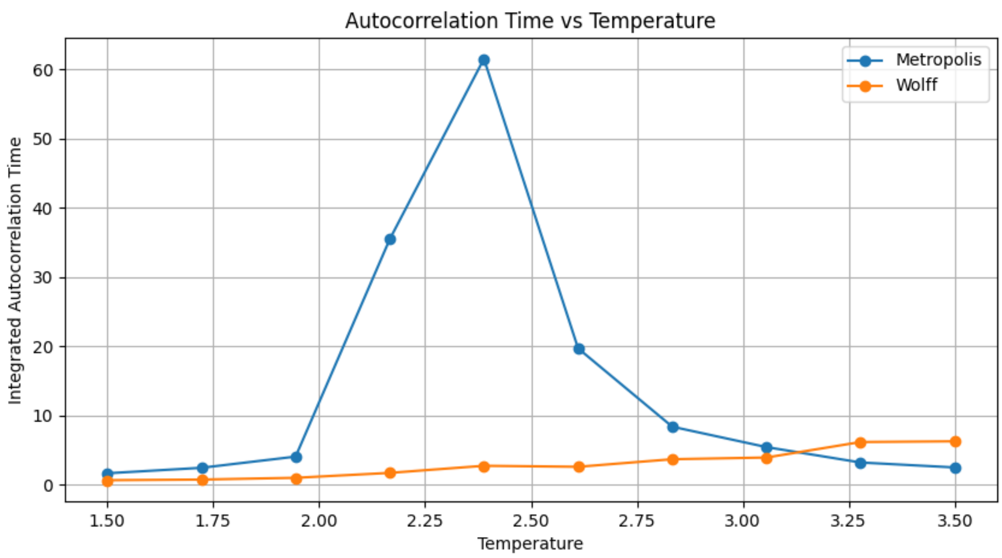


Figure 7: Autocorrelation time τ vs Temperature. Note the divergence of τ for the SSF algorithm near T_c, contrasted with the suppression of critical slowing down in the cluster algorithm.

4.2.3 Specific Heat

The specific heat capacity exhibits a logarithmic divergence near the critical temperature. The peak location provides a finite-size estimate for T_c.

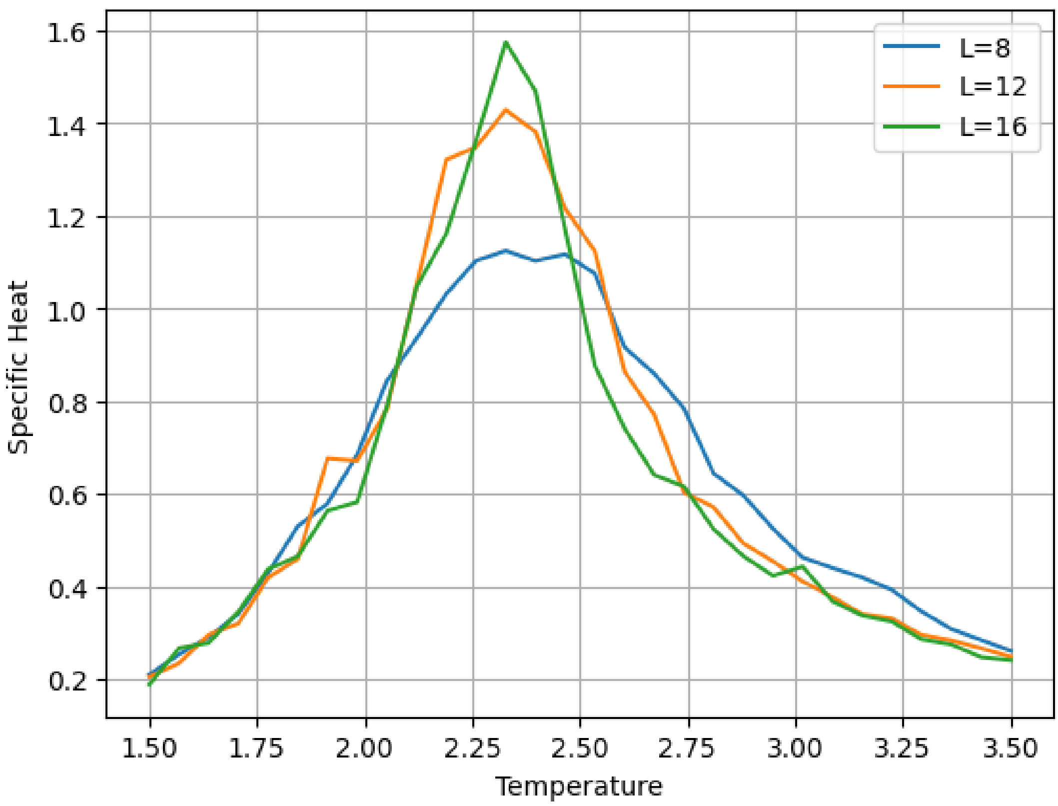


Figure 8: Specific Heat C_V vs Temperature. The peaks align near the theoretical critical temperature $T_c \approx 2.269$.

5 Advanced Topics

5.1 The XY Model

We extended our Monte Carlo study to the 2D XY model, where the discrete Ising spins are replaced by classical unit vectors $\vec{s}_i = (\cos\theta_i, \sin\theta_i)$ possessing continuous $O(2)$ rotational symmetry. The Hamiltonian is given by:

$$H = -J \sum_{\langle i,j \rangle} \vec{s}_i \cdot \vec{s}_j = -J \sum_{\langle i,j \rangle} \cos(\theta_i - \theta_j)$$

To simulate this efficiently, we adapted the Wolff cluster algorithm. In each step, a random reflection vector $\hat{r}$ is chosen and spins are reflected ($\vec{s}_i \to \vec{s}_i - 2(\vec{s}_i \cdot \hat{r})\hat{r}$) based on the bond activation probability $P_{add} = 1 - \exp(-2\beta(\vec{s}_i \cdot \hat{r})(\vec{s}_j \cdot \hat{r}))$. The results reveal fundamental differences from the Ising universality class:

1. **Absence of Magnetization:** Consistent with the Mermin-Wagner theorem, we observed no spontaneous magnetization at non-zero temperatures. Continuous symmetries cannot be broken in 2D at finite T, preventing true long-range order.

2. **Topological Transition:** Despite the lack of magnetic ordering, the system undergoes a Kosterlitz-Thouless (KT) transition. This is driven by the unbinding of topological defects (vortex-antivortex pairs) rather than symmetry breaking.

3. **Specific Heat Behavior:** The specific heat C_V does not exhibit the sharp, divergent peak characteristic of the Ising model. Instead, it shows a broader, non-divergent peak located slightly above the actual transition temperature T_{KT}. This confirms that the energy fluctuations associated with vortex unbinding are distinct from those in second-order magnetic transitions.

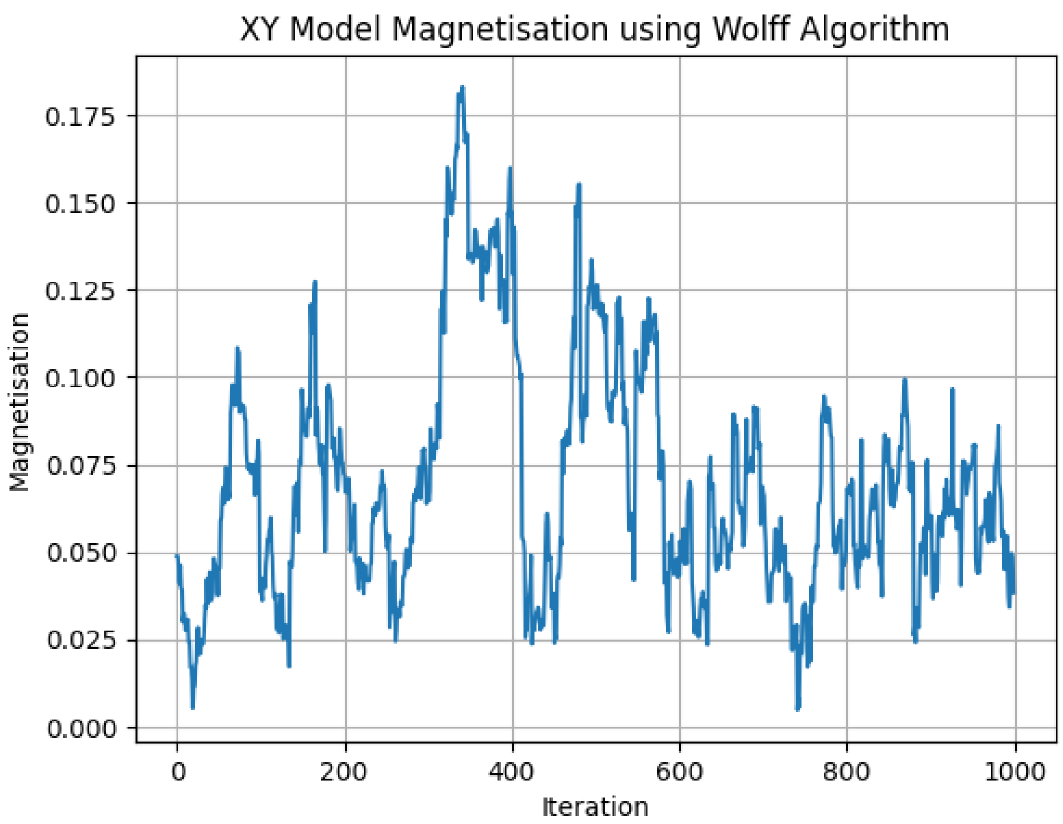


Figure 9: Specific Heat vs Temperature for the XY Model.

5.2 Unsupervised Learning: PCA

To investigate the phase transition without the manual construction of an order parameter, we applied Principal Component Analysis (PCA) to the raw lattice configurations. PCA is a dimensionality reduction technique that identifies the directions (principal components) of maximum variance in the high-dimensional configuration space.

The methodology was executed as follows:

1. **Data Acquisition:** We generated configurations for a spectrum of temperatures ranging from $T = 1.6$ to 3.0. For each temperature, the system was evolved through 500 Swendsen-Wang or Wolff cluster flips to ensure thermal equilibration.

2. **Preprocessing:** Each $L \times L$ grid was flattened into a $1 \times L^2$ vector. These vectors were concatenated into a comprehensive data matrix, with associated temperature values stored for post-simulation labeling.

3. **Dimensionality Reduction:** The first and second principal components were extracted from the dataset using the `sklearn.decomposition.PCA` module.

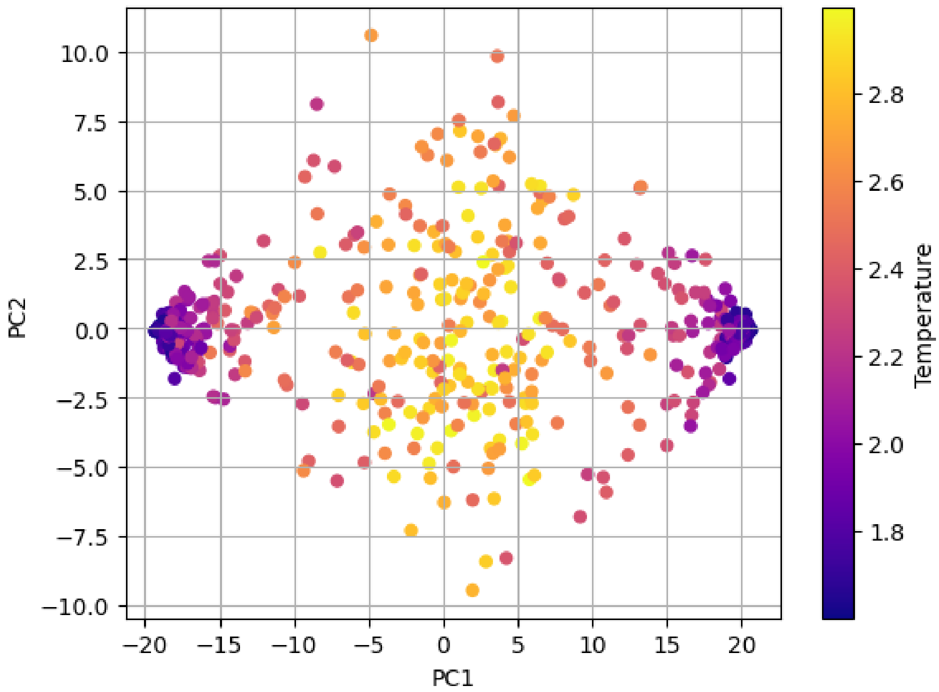


Figure 10: PCA projection of Ising configurations. The data clearly separates into three clusters: Ordered Up (Low T), Ordered Down (Low T), and Disordered (High T), demonstrating unsupervised detection of the phase transition.

The resulting visualization (Figure 10) provides significant physical insights:

- **Phase Separation:** The scatter plot reveals a clear bifurcation between high-temperature and low-temperature states. High-temperature configurations cluster near the origin, representing the disordered paramagnetic phase.
- **Symmetry Breaking:** At low temperatures, the configurations separate into two distinct clusters corresponding to the magnetized states ($\sigma \approx +1$ and $\sigma \approx -1$).
- **Physical Interpretation:** The first principal component (PC1) typically aligns with the average magnetization of the system, effectively "discovering" the order parameter of the Ising model in an unsupervised manner.

This result demonstrates that the essential physics of the phase transition—specifically the emergence of order and the breaking of $\mathbb{Z}_2$ symmetry is encoded directly in the variance of the raw configuration data.

The first principal component (PC1) strongly correlates with the magnetization order parameter, effectively "learning" the physics of the transition without prior labeling.

6 Conclusion

In this project, we performed a comprehensive Monte Carlo study of the two-dimensional Ising model, successfully replicating its phase transition and critical behavior.

Our comparison of simulation algorithms demonstrated the severe limitations of local update methods near criticality. The Single Spin Flip (Metropolis) algorithm exhibited a diverging autocorrelation time τ near the critical temperature T_c, a phenomenon known as critical slowing down. In contrast, the cluster-based Wolff algorithm effectively suppressed this divergence by flipping large domains of correlated spins simultaneously, proving to be indispensable for efficient simulation in the critical region.

Thermodynamic analysis of the magnetization and specific heat capacity revealed a continuous second-order phase transition. As the lattice size L increased, the transition sharpened, highlighting the role of finite-size effects. From the divergence of the specific heat and the inflection point of the magnetization, we estimated the critical temperature to be $T_c \approx 2.34$ (in units of J/k_B). This numerical estimate is in excellent agreement with the exact theoretical value provided by Onsager's solution ($T_c \approx 2.269$).

Furthermore, our investigation into advanced topics yielded significant insights:

- **The XY Model:** Extensions to continuous $O(2)$ symmetry showed a distinct lack of spontaneous magnetization, consistent with the Mermin-Wagner theorem. The specific heat data suggested a topological Kosterlitz-Thouless (KT) transition rather than a standard symmetry-breaking transition.

- **Unsupervised Learning:** The application of Principal Component Analysis (PCA) successfully distinguished between ordered and disordered phases without any prior knowledge of the order parameter. The first principal component was found to correlate strongly with magnetization, validating the use of machine learning techniques for detecting phase transitions in many-body physics.

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